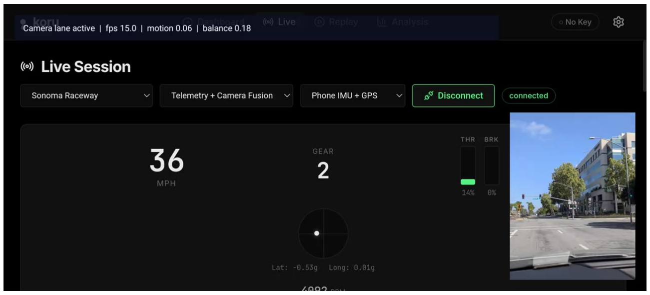
A Domain Expertise Layer on top of the Google stack

Today's best telemetry coaches (Garmin Catalyst) tell you what went wrong with numbers, after the fact. This system delivers context-aware coaching *as it happens*, adapted to driver skill, by routing every decision through a layered domain doctrine — Ross Bentley pedagogy, T-Rod's Sonoma session notes, and corner-by-corner Sonoma doctrine — before any LLM ever speaks. **Gemini becomes a tool inside a trust framework, not the source of trust.**

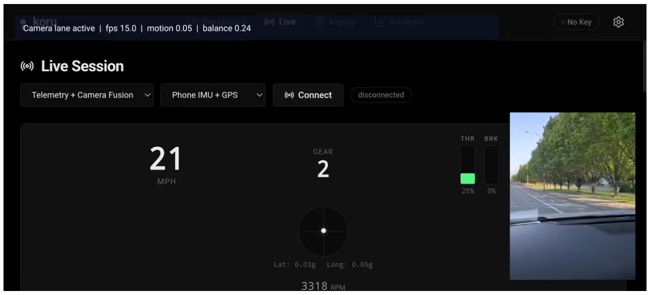
Owner	Branch	Tests	Stack
Team 1	consolidated	60 / 60 passing (13 files)	Gemma 4 E2B-it (LiteRT-LM) · Gemini 3.1 Pro / 3 Flash preview Gemini 2.5 Flash Lite · Gemini 2.5 Pro TTS · Pixel 10 · CameraX 1.4.2 · MediaPipe Tasks GenAI 0.10.27

Live demo — captured on Pixel 10

Two screen captures from the consolidated build. Telemetry is the master clock; the latest CameraX vision frame is fused onto each telemetry frame; the DEL shapes everything that comes out the speaker. Embedded MP4 versions ship in the design-assets folder alongside this PDF.



Demo 1 · Sonoma Raceway · Telemetry + Camera Fusion
54s · Phone IMU + GPS source · 36 mph, Gear 2, Lat -0.53g during a left-hander. Vision lane active (fps 15, motion 0.06).
`media/demo-sonoma-live.mp4`



Demo 2 · Device Camera + GPS Test → Telemetry + Camera Fusion
2m21s · Trackless device test transitioning to fused mode at 21 mph. Validates the camera pathway independently from telemetry. `media/demo-camera-fusion.mp4`

Three pillars of the framework

1. Domain Expertise Layer (load-bearing)	2. Split-Brain Coaching Engine	3. Trust Gates Before Speech
Ross Bentley's 7 mental models + T-Rod's 5 Sonoma coaching patterns + corner-by-corner Sonoma doctrine. Not prompt text — it actively suppresses premature advice, biases hot-path wording, and rewrites feedforward per corner and per skill level. Doctrine, not decoration.	Hot path (12-rule DecisionMatrix, <50ms, no I/O). Cold path (`gemini-2.5-flash-lite`, 2–5s). Edge reasoner cascade: AICore (Gemma 4 preview) → LiteRT-LM (Gemma 4 E2B-it) → Deterministic. The hot path NEVER waits for the cold path. P0 safety preempts everything.	Timing state machine (OPEN → DELIVERING → COOLDOWN → BLACKOUT during apex). Driver model classifies skill from input smoothness. Bench-safe motion gating suppresses stationary false positives. Confidence thresholds gate every utterance.

Verified tech stack — exact models, exact versions

Pulled directly from `pixel-android-app/app/build.gradle.kts`, `models/current-model.json`, `useGeminiCloud.ts`, `geminiService.ts`, and `coachingService.ts`.

Layer	Component	Exact identifier in code
On-device LLM · tier 1	**Gemini Nano 4 / Gemma 4 preview** via Android AICore Prompt API	`com.google.ai.edge.aicore.GenerativeModel` `runtime/reasoner/AiCoreReasoner.kt`

On-device LLM · tier 2 (active)	**Gemma 4 E2B-it** via LiteRT-LM (MediaPipe Tasks GenAI)	`litert-community/gemma-4-E2B-it-litert-lm`
		`file: gemma-4-E2B-it.litertlm`
		`sha256: ab7838cd…b27e42 (verified)`
		`device path: /data/local/tmp/koru/models/gemma-4-e2b-it/`

On-device LLM · tier 3 (floor)	**Deterministic-only** — phrase catalog + 12-rule DecisionMatrix, no LLM	`runtime/reasoner/DeterministicOnlyReasoner.kt` `guarantees the system never goes silent on ML failure`

Cloud cold path	**gemini-2.5-flash-lite** via REST v1beta	`https://generativelanguage.googleapis.com/v1beta/models/`
		`gemini-2.5-flash-lite:generateContent`
		`services/coachingService.ts:650`

Post-session (deep)	**gemini-3.1-pro-preview** with ThinkingLevel.HIGH	`hooks/useGeminiCloud.ts:14, 99–104` `via @google/genai 1.45.0`
Post-session (fast)	**gemini-3-flash-preview** with ThinkingLevel.LOW	`hooks/useGeminiCloud.ts:15, 99–104`

TTS	**gemini-2.5-pro-preview-tts** · @google/genai v1alpha streaming	`responseModalities: [Modality.AUDIO]`
		`hooks/useGeminiCloud.ts:124–127,`
		`useTTS.ts:83`

Android	Kotlin · MediaPipe Tasks GenAI 0.10.27 · CameraX 1.4.2	`minSdk 29 · target 35 · JVM 17`
		`com.google.mediapipe:tasks-genai:0.10.27`
		`androidx.camera:camera-* 1.4.2 · webkit 1.12.1`

Web	React 19.2 · TypeScript 5.9 · Vite 8 · Tailwind 4 · Recharts · Papaparse	`@google/genai 1.45.0 · Vitest 4.1` `13 test files · 60 / 60 passing`

Three-tier on-device reasoner cascade. `EdgeRuntimeManager.warmupPreferredBackend()` probes each backend in order. The first one that responds with a non-UNAVAILABLE / non-ERROR state becomes the active reasoner; the choice is persisted to `SharedPreferences`. Confidence per tier: AICore **0.78**, LiteRT-LM **0.72**, Deterministic **baseline**. The deterministic floor is what makes this trustable — the system never goes silent because of an ML failure.

Tier 1 · AICore	Tier 2 · LiteRT-LM	Tier 3 · Deterministic
Gemini Nano 4 / Gemma 4 preview Prompt API. Probed first via reflection on `com.google.ai.edge.aicore.GenerativeModel`. Highest confidence (0.78). Supports HOT, FEEDFORWARD, EDGE paths.	Gemma 4 E2B-it (.litertlm, sha256-verified). Loaded via MediaPipe `tasks.genai.llminference.LlmInference`. Pushed to device by `npm run pixel:e2e:prepare`.	Phrase catalog + 12-rule decision matrix. No LLM. Always available. This is the trustable floor — the system never goes silent due to ML failure.

Domain Expertise Layer — what's actually in it

Implemented in `src/data/trackExpertise.ts`, `src/data/trodCoachingData.ts`, and `src/utils/coachingKnowledge.ts`. Wired into the runtime selector, not just the prompt.

Pedagogy / Doctrine source	What it contributes at runtime
Ross Bentley pedagogy (7 mental models) Friction Circle · Weight Transfer · Trail Braking · Vision Drives the Car · Maintenance Throttle · Slow In Fast Out · One Thing at a Time	Skill-adapted humanization, beginner trigger phrases ('Hard initial!', 'Squeeze, don't stab', 'Eyes up!'), session-progression suppression of advanced techniques in the first 60s.
T-Rod Sonoma coaching (5 patterns) Throttle Commitment · Brake Trace Quality · Delay Early Throttle · Distance Is King · Session Learning Sequence	Hot-path rules (EARLY_THROTTLE, LIFT_MID_CORNER, SPIKE_BRAKE) and corner-specific feedforward wording on Sonoma sweepers.
Sonoma track doctrine Corner dossiers for T2, T3, T3A, T6, T7, T9–10, T11, T12; three timing sectors with distinct guidance.	Feedforward triggers tied to corner geometry; sector-specific advice replaces generic LLM output. Sonoma is the primary deployed track in both web registry and native Android catalog.
Driver model + session goals 10s rolling window, 5s hysteresis, max 3 pre-race goals.	Beginner: feel-based feedback. Intermediate: persona phrasing. Advanced: data-driven (G-Lat, exit speed delta). Goals bias hot-path rule activation.

> *“The right message at the wrong time is the wrong message. Feedback 800ms late is worse than silence.” — Brian Luc*

Split-Brain Coaching Engine

Path	Latency	Backed by	Tier	Role
Hot	< 50 ms	12-rule DecisionMatrix + DriverModel	P0/P1/P3	Per-frame act-now coaching. Rules: OVERSTEER_RECOVERY, THRESHOLD, TRAIL_BRAKE, COMMIT, THROTTLE, PUSH, COAST, HESITATION, FULL_THROTTLE, EARLY_THROTTLE, LIFT_MID_CORNER, SPIKE_BRAKE.
Cold	2–5 s	`gemini-2.5-flash-lite` via REST v1beta, skill-adapted prompt	P2	Multi-frame physics analysis. 10 / 20 / 15-word ceiling per skill. Enriches future hot-path; never blocks.
Feedforward	trigger	Geofence + Sonoma T-Rod corner dossiers	P1	Anticipatory advice 150m before corner. Per-corner wording from T-Rod notes.
Edge	on-device	AICore (Gemma 4 prev) → LiteRT-LM (Gemma 4 E2B-it) → Deterministic	P1/P2	Three-tier cascade. Auto-warms; persisted to SharedPreferences.

Post	5–30 s	`gemini-3.1-pro-preview` (HIGH) · `gemini-3-flash-preview` (LOW)	review	Sector-by-sector lap analysis with explicit ThinkingLevel. Runs on saved artifact, not live.
TTS	50–200 ms	`gemini-2.5-pro-preview-tts` · @google/genai v1alpha	output	Persona voice. generateContentStream + Modality.AUDIO. Web Speech API fallback.

Trust gates that fire before any utterance

Gate	Status	What it does
Timing state machine	**Done**	OPEN→DELIVERING→COOLDOWN→BLACKOUT. Beginners: 3s cooldown, mid-corner blackout. Advanced: 1s, no blackout. P0 bypasses all.
Priority queue + preempt	**Done**	P0 preempts all. Max 5 items, 3s stale expiry. Safety alerts never sit behind technique tips.
Driver model + hysteresis	**Done**	10s rolling window (rate-agnostic 8–25Hz). 5s hysteresis. Smoothness < 0.4 OR coasting > 30% ⇒ BEGINNER.
Bench-safe motion gating	**Done**	Phone-only telemetry coaching suppressed until real movement detected. Eliminates stationary false positives.
Session phase filter	**Done**	Phase 1 (0–60s) suppresses TRAIL_BRAKE, COMMIT, ROTATE, EARLY_THROTTLE. Advanced drivers skip to Phase 3.
Confidence + speed gates	**Done**	Realtime speech hardening. Higher-priority cues interrupt lower-priority speech. Faster timing for advanced.
Pre-rendered safety MP3s	**Planned**	AudioContext pre-caching wired; need per-persona BRAKE/OVERSTEER clips to bypass TTS latency.
Cold path offline fallback	**Future**	Pre-computed coaching lookup table per known track. Pixel 10 + Gemma 4 evaluated as Nano upgrade.

How the Google stack plugs in

Layer	Google component	Exact identifier in code	Status
Compute / OS	Pixel 10 · Android (minSdk 29, target 35)	KoruTelemetryService foreground service · JVM 17	Done
Vision	CameraX 1.4.2	camera/CameraLaneAnalyzer.kt · VisionFeatureStore.kt	Done
On-device runtime	MediaPipe Tasks GenAI 0.10.27	com.google.mediapipe:tasks-genai:0.10.27 · LlmInference	Done
On-device LLM (preferred)	Gemini Nano 4 / Gemma 4 preview · AICore	com.google.ai.edge.aicore.GenerativeModel	Scaffolded
On-device LLM (active)	Gemma 4 E2B-it via LiteRT-LM	litert-community/gemma-4-E2B-it-litert-lm · sha256-verified	Done
Cold path	gemini-2.5-flash-lite	v1beta REST · coachingService.ts:650 · geminiService.ts:30	Done
Post-session deep	gemini-3.1-pro-preview, ThinkingLevel.HIGH	hooks/useGeminiCloud.ts:14 · @google/genai 1.45.0	Done
Post-session fast	gemini-3-flash-preview, ThinkingLevel.LOW	hooks/useGeminiCloud.ts:15	Done
TTS	gemini-2.5-pro-preview-tts + Web Speech API	generateContentStream · Modality.AUDIO · v1alpha	Done
Storage	BigQuery / GCS (AGY pipeline)	Future post-session schema for cross-session learning	Planned

Latency budget — telemetry event to spoken coaching

Stage	Latency	Notes
RaceBox BLE → Pixel 10	7–15 ms	BLE 5.2 high-priority connection
OBD K-Line round-trip	80–150 ms	Vehicle-side, cannot reduce
Android BLE → fusion	5–10 ms	Foreground service
AI inference (hot path)	< 50 ms	Heuristic rules, no I/O
TTS audio output	50–200 ms	Web Speech / Gemini TTS
TOTAL BUDGET	**~200–425 ms**	Under Brian Luc's 800ms 'worse than silence' threshold

Cross-pollination opportunities for the cohort

These are the patterns we believe should be lifted to cohort-level standards. The DEL shape transfers to any domain (medical, ops, fraud); the racing specifics don't.

Opportunity	Status	Why it matters across teams
VehicleDataStream contract	**Proposed**	Abstract OBD/CAN/BLE behind one interface. Coaching engine never changes when upgrading from OBD to direct CAN. Ready for Team 2 (CAN-to-USB) and Team 3 (MoTec).
Shared post-session schema	**Open**	Per-corner metrics, mistake zones, coaching decisions. Codify the saved-session artifact format so AGY pipeline ingestion is identical across the three teams.
Priority tier convention	**Reusable**	P0 safety / P1 tactical / P2 strategic / P3 encouragement. Trivially portable. Recommended cohort default for any real-time advisory work.
Time-based windows over frame counts	**Reusable**	Sensor rates differ (8 Hz OBD vs 25 Hz RaceBox). Always store {time, value} pairs and trim by elapsed time. One decision eliminated an entire bug class.
Domain Expertise Layer pattern	**Reusable**	Doctrine that filters and shapes LLM output is the trust mechanism. Required between any raw model and any user-facing output.
Mocked telemetry endpoint	**In progress**	Throttled synthetic stream so all teams validate ingestion pre-field-test. Owned by Team 1 — adopting it unblocks pipeline development for everyone.

Repository & supporting documents

Asset	Link / Path
GitHub — Team 1 primary repository ('consolidated' branch)	`https://github.com/rabimba/trustable-ai-codelab`
GitHub — upstream cohort reference (ykro)	`https://github.com/ykro/trustable-ai-codelab`
Interactive design asset (HTML, embedded video demos)	`design-assets/trustable-ai-framework.html`
Demo video 1 — Sonoma Live · Telemetry + Camera Fusion	`design-assets/media/demo-sonoma-live.mp4`
Demo video 2 — Device Camera + GPS Test → Fusion	`design-assets/media/demo-camera-fusion.mp4`
End-to-End Architecture	`docs/end-to-end-architecture.md`
Data Reasoning module reference	`docs/data-reasoning.md`
Real-Time Reasoning Learnings (portable patterns)	`docs/learnings-real-time-data-reasoning.md`
Consolidation Report (what was merged from where, attribution)	`docs/consolidation-report.md`
Implementation history	`docs/implementation-history.md`
Current implementation audit	`docs/current-implementation-audit.md`
User stories	`docs/user-stories.md`
Pre-race chat contract	`docs/pre-race-chat-contract.md`

Attribution. All implementation, integration, and consolidation work in this branch is attributed to **Team 1**. Implementation, integration, architecture, and documentation by **Rabimba Karanjai**. Imported reference artifacts retain their original attribution: the hot-path latency benchmark and browser/data-reasoning reference architecture trace to Adrian Catalan; the cross-team user-story framing traces to Madona S Wambua. Domain coaching content is sourced from Ross Bentley, T-Rod (Tony Rodriguez), and Brian Luc and is integrated as doctrine, not modified.

Built with the Google stack · Pixel 10 · CameraX · Gemini · LiteRT-LM